(A-24) Seat No.:________ No. of Printed Pages : 2

SARDAR PATEL UNIVERSITY

F.Y.B.C.A. (Semester - II) (2010 Batch) EXAMINATION March – 2017 (NC)

US02CBCA01 : Advanced C Programming and Introduction to Data Structures

Date:18/03/2017, Saturday Time: 02:00 to 05:00 Total Marks : 70

Q.1 Multiple choice questions: [10]

1. Which of the following is not a C memory allocation function?
   (a) malloc() (b) calloc()
   (c) realloc() (d) alloc()
2. Which of the following is not a derived data type?
   (a) Arrays (b) Float
   (c) Structure (d) Pointers
3. Which of the following can be used to create a new type that can be used anywhere a type is permitted?
   (a) typedef (b) array
   (c) struct (d) Both struct and typedef
4. Which one of the following is valid for opening a file for only reading?
   (a) fileOpen (filenm, "r"); (b) fileOpen (filenm, "ra");
   (c) fopen (filenm, "r"); (d) fopen (filenm, "read");
5. The term "push" and "pop" is related to the?
   (a) array (b) queue
   (c) stacks (d) All of these
6. Two dimensional arrays are also called?
   (a) tables arrays (b) matrix arrays
   (c) both A and B (d) None of these
7. A data structure that contains not only a data field but also contains pointer field is known as __________.
   (a) Queue (b) Tree
   (c) Stack (d) Linked List
8. What are two predefined FILE pointers in C?
   (a) stdout and stderr (b) console and error
   (c) stdio and stderr (d) stdout and stdio
9. A data structure in which insertion and deletion of an elements occurs at only one end is known as ________.
   (a) Queue (b) Stack
   (c) Tree (d) Graph
10. Which of the following statement is FALSE for the Queue data structure?
   (a) Its nature is LIFO (b) Its nature is FIFO
   (c) It is a non- primitive data structure (d) It is a Linear data structure

(P.T.O.)

Q.2 Attempt any six out of eight. [12]

1. List out operations that can be performed on pointers.
2. Differentiate '.' and '->' operators.
3. Differentiate: getc and getchar
4. Explain the fclose() function with example.
5. What do you mean Linear Data Structure?
6. List out different applications of data Structure.
7. State various Applications of Linked List.
8. Define: Queue and Deque.

Q.3 (a) Write a note on Dynamic memory allocation. [5]
(b) Write note on: pointer to pointer [3]

OR

Q.3(a) Define pointer variable. How can we declare and initialize pointer variable? How can we access value of variable through pointer type variable? [5]
(b) Explain pointer to structure using suitable example. [3]

Q.4 (a) What is union? Explain its definition, declaration and assigning values to members of union. [4]
(b) Explain pointer to structure array using appropriate example. [4]

OR

Q.4(a) What is structure? Explain its definition, declaration and assigning values to members of structure. [4]
(b) Write note on: structure within structure [4]

Q.5 Explain the all the modes of file management with example. [8]

OR

Q.5 Explain the getc, putc, getw and putw function with example. [8]

Q.6 (a) Explain the data structure with c briefly. [4]
(b) Write a short note on primitive data structure operations. [4]

OR

Q.6 (a) Explain the linear and non linear data structure briefly. [4]
(b) Write down advantages of data structure. [4]

Q.7 (a) Write an algorithm to insert an element at the beginning of a Singly linked list. [4]
(b) Write a short note on Singly linked List. [4]

OR

Q.7(a) Write an algorithm to insert an element at the ending of a Singly linked list. [4]
(b) What is linked list? Explain types of linked list. [4]

Q.8 (a) Explain a STACK with an example of various operations. [6]
(b) Define: Circular Queue and Priority Queue. [2]

OR

Q.8(a) Explain a QUEUE with an example of various operations. [6]
(b) Differentiate: peep and change operation. [2]

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